

AIMO Proof Pilot — Final Grade

Problem (#_ID): 6_AMAZIN

Submission: team_4

Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (docs/markschemes.pdf).

1. Final mark

Final mark (choose one):

2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. **ME1**) where possible.

Used Cartesian coordinates and split into two cases corresponding to the ones in the mark scheme. The correct Pell's equation was derived, but $n = 3$ wasn't excluded, resulting in a sequence which is off by one index. The script satisfies NEC2 which means it cannot be 6/7 marks. Instead, it obtains PP2 which offers a mark.